Comprehensive Analysis of DBSCAN and KMeans Clustering for Anomaly Detection in Weekly Sales Transaction Data

1. Introduction

Anomaly detection, also known as outlier detection, using DBSCAN is a powerful method for identifying unusual patterns in data, making it essential for recognizing errors and significant deviations. Using techniques like KMeans clustering, we can effectively group similar data points, which enhances our analysis of sales trends in the **UCI**'s **Sales Transactions Dataset**, comprising **811 products** over **52 weeks**. This dataset offers crucial insights into typical sales behaviours, seasonal patterns, and potential anomalies that clearly indicate stock issues or data errors.

2. Preprocessing

The data preprocessing steps began with an examination of the dataset, which consists of 811 products over 52 weeks, ensuring no **missing values** or **duplicates**. We extracted the weekly sales columns as **features** while retaining product codes as **identifiers**, applied **VarianceThreshold** to remove **near-zero variance features**, and utilized **RobustScaler** for **normalization** focused on outlier-resistant statistics. Finally, we implemented **Principal Component Analysis** (**see Appendix 1**) to **reduce dimensionality** while preserving **95% of the variance**, thereby creating an optimized foundation for clustering algorithms with a **standardized scale** and improved **resilience** to outliers.

3. Data Analysis

The **first histogram** (**see Appendix 2**), "Total Product Sales Per Week", shows weekly sales for all products throughout the year, ranging from about 500 to 8,500 units. Sales are mostly steady, but there is a noticeable dip around Week 26, followed by a slow recovery. This decline may suggest a seasonal effect or an outside factor. The **second histogram** (**see Appendix 3**), "Distribution of Total Sales Per Product", shows total sales for different products over 52 weeks. The distribution is **right-skewed**, meaning most products sold few units (below 500 units over 52 weeks), while a few top products sold up to the range of 2,500 units. This highlights a long-tail effect where the best sellers drive total sales, providing useful insights for inventory management and marketing.

4. DBSCAN Implementation

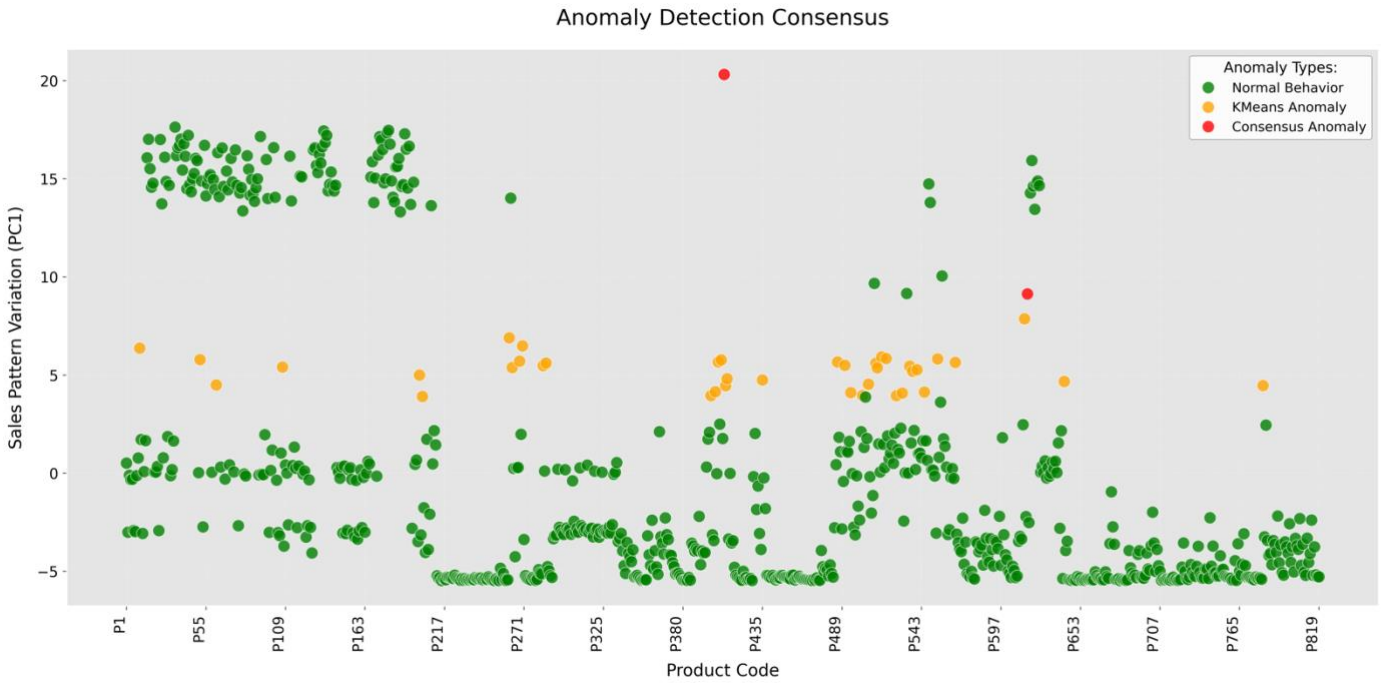
The DBSCAN algorithm was implemented to PCA-reduced sales data, retaining **95% variance** to categorise product groupings and outliers, with **ε=3.77** automatically selected through **k-distance graph** (**see Appendix 4**) analysis and **min_samples** set to twice the feature count. The clustering illustrates (**see Appendix 5**) two distinct groups: **Cluster 1** (689 products) and **Cluster 2** (117 products), along with 5 anomalous products (P208, P262, P409, P557, P615) exhibiting irregular weekly sales patterns, for example, P409 showed consistently elevated sales across weeks 0 - 3. The results depict effectiveness of DBSCAN in identifying natural clusters while highlighting true outliers for further investigation.

5. KMeans Implementation

A KMeans clustering analysis was executed on PCA reduced sales data, retaining **95% variance**, to recognise product groupings. The **Elbow Method** (**see Appendix 6**) suggested an optimal cluster number of **k=3**, while the **Silhouette Score** (**see Appendix 7**) peaked at **k=2** with a score of **0.783** recommended two clusters as ideal for cohesion and separation. The clustering plot (**see Appendix 8**) showed two distinct groups: **Cluster 1** and **Cluster 2** along with **centroids**. These results aligned with DBSCAN findings, which also classified two main clusters, however, K-means forms stricter spherical boundaries.

6. Comparison

A **consolidated anomaly detection approach** combining KMeans (distance-to-centroid) and DBSCAN (density-based) methods was applied to PCA reduced sales data which resulted in the classification of products into three categories: (1)**Consensus Anomaly**: flagged by both methods, (2) **KMeans Anomaly**: flagged only by K-means since, a product with steady mid-range sales might be centrally located in a DBSCAN cluster however, far from the KMeans centroid due to the spherical cluster assumption of the KMeans, (3)**Normal Behaviour**: not flagged as outliers by either DBSCAN or KMeans. The Consensus plot flags products with the most notable sales pattern deviations such as, erratic weekly trends, as highlighted by both KMeans and DBSCAN. This dual-method validation **reduces false positives** by requiring outlier agreement from two distinct clustering perspectives, securing robust anomaly detection.



7. Evaluation Metrics

The evaluation metrics and runtime reveal key differences between the DBSCAN and KMeans clustering algorithms. DBSCAN demonstrated superior efficiency, with a training time of **0.0099 seconds** and near-instantaneous inference. It achieved high accuracy (**0.9951**), specificity (**0.9951**), and ROC AUC (**0.9975**), though with moderate precision (**0.2000**) and F1 score (**0.3333**), suggesting some false positives (**see Appendix 9**). However, its perfect recall (**1.0000**) confirms that it successfully identified all true anomalies. Conversely, KMeans required longer training (**0.0353 seconds**) and inference (**0.0019 seconds**). While it also achieved perfect recall (**1.0000**) and a strong ROC AUC (**0.9716**), its extremely low precision (**0.0244**) resulted in approximately **40 false alerts per true anomaly**, and its F1 score (**0.0476**) reflected this imbalance. Despite reasonable accuracy (**0.9507**) and specificity (**0.9506**), these results underscore the inherent precision-recall trade-off in distance-based anomaly detection. In summary, DBSCAN excels in speed and practical deployment for real-time applications, whereas both methods ensure comprehensive detection of true anomalies.

Metric	DBSCAN	KMeans
Accuracy	0.9951	0.9507
Precision	0.2000	0.0244
Recall	1.0000	1.0000
F1-Score	0.3333	0.0476
ROC AUC	0.9975	0.9716
Specificity	0.9951	0.9506
Training Time	0.0099 seconds	0.0353 seconds
Inference Time	0.0000 seconds	0.0019 seconds

8. Conclusion

This analysis emphasizes the **complementary** roles of DBSCAN and KMeans in sales anomaly detection. **DBSCAN** is particularly **well-suited** for most operational scenarios, demonstrating high computational efficiency (0.0099 seconds training time) and strong detection performance (ROC AUC=0.9975, accuracy=0.9951). Despite some false positives (precision=0.2000), it reliably identifies all true anomalies (recall=1.0000) making it suitable for real-time monitoring. On the other hand, **KMeans** achieves perfect recall but has a significant false-positive ratio of 40:1 due to its spherical cluster assumption, making it viable only for **high-risk situations** where missing anomalies is unacceptable such as in fraud detection. The methods have distinct **limitations** that are worth noting: DBSCAN is less effective when faced with non-uniform data density distributions, while KMeans encounters significant challenges due to its assumption of spherical clusters, particularly in high-dimensional feature spaces. To conclude, **DBSCAN provides an effective balance** of speed and accuracy for sales analytics, while **KMeans** can serve as a **complementary screening system**. These findings offer valuable guidance for selecting anomaly detection methods tailored to specific business needs.

9. References

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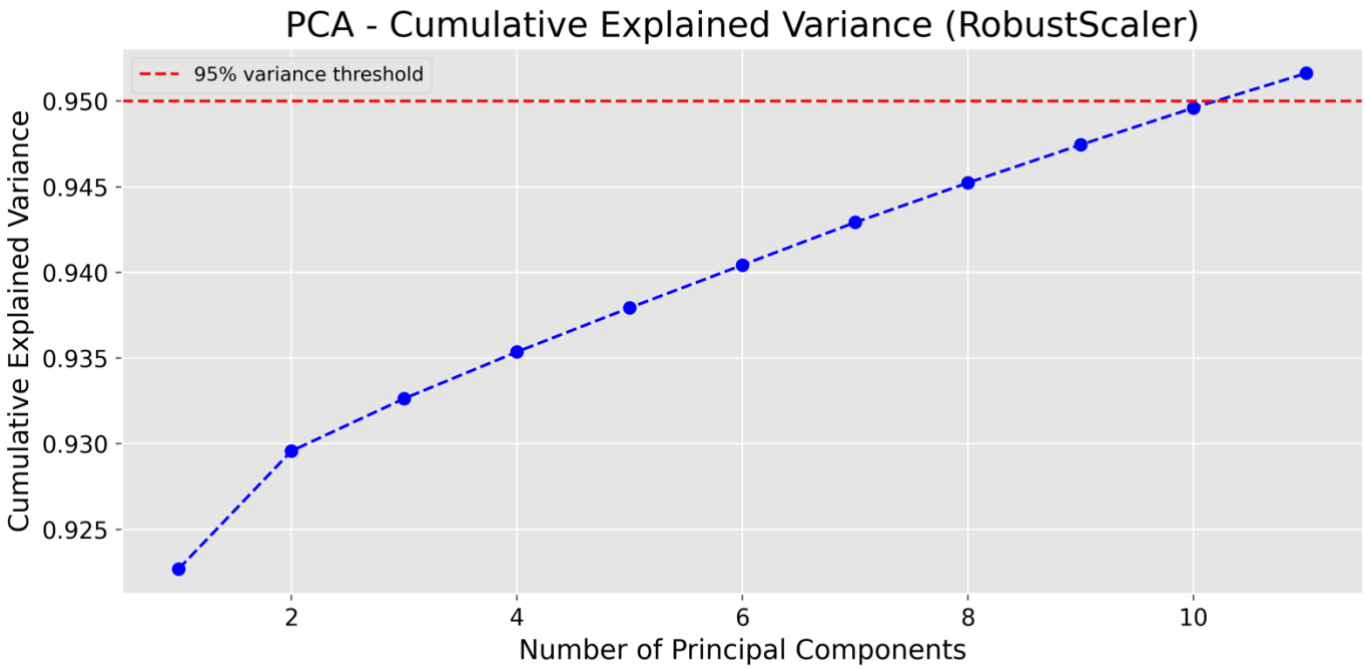
UCI Machine Learning Repository (2015) *Sales transactions dataset weekly*. Available at: https://archive.ics.uci.edu/dataset/396/sales+transactions+dataset+weekly (Accessed: 1 April 2025).

10. Appendices

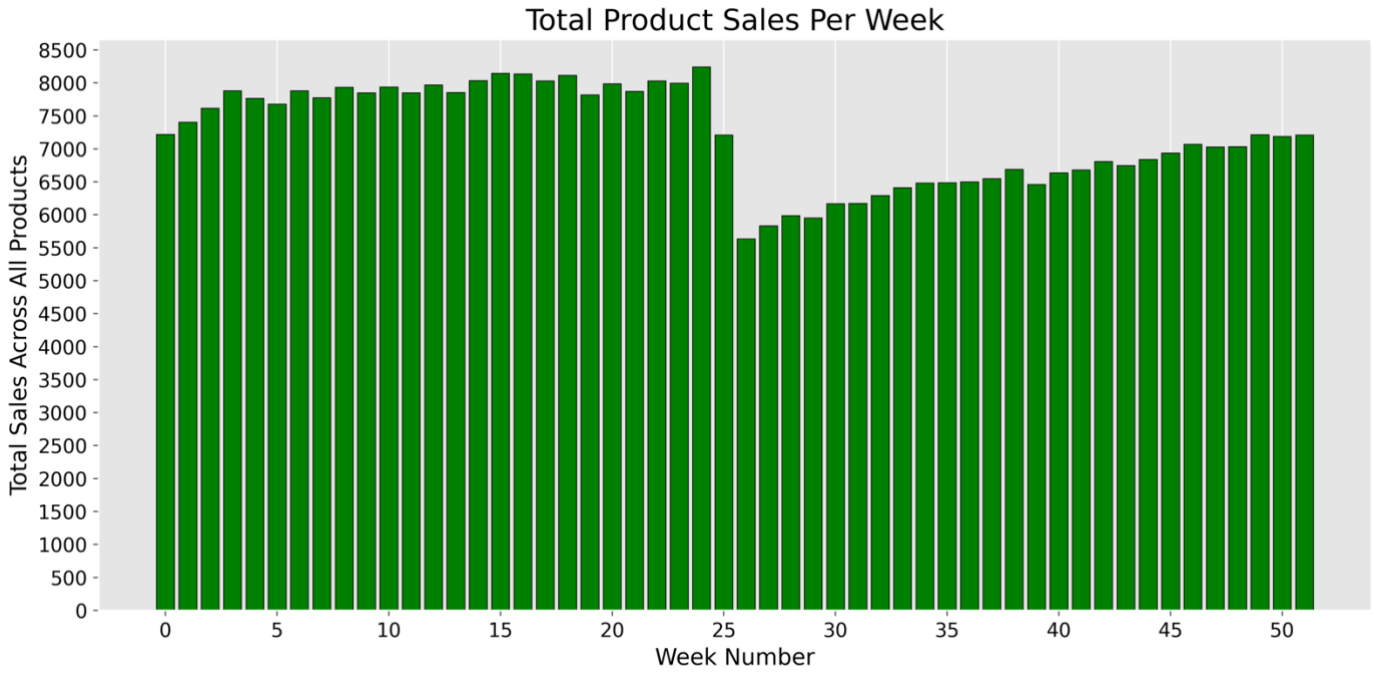
Appendix 1: PCA - Cumulative Explained Variance (RobustScaler) Analysis

This plot shows the cumulative variance explained by principal components after RobustScaler normalization.



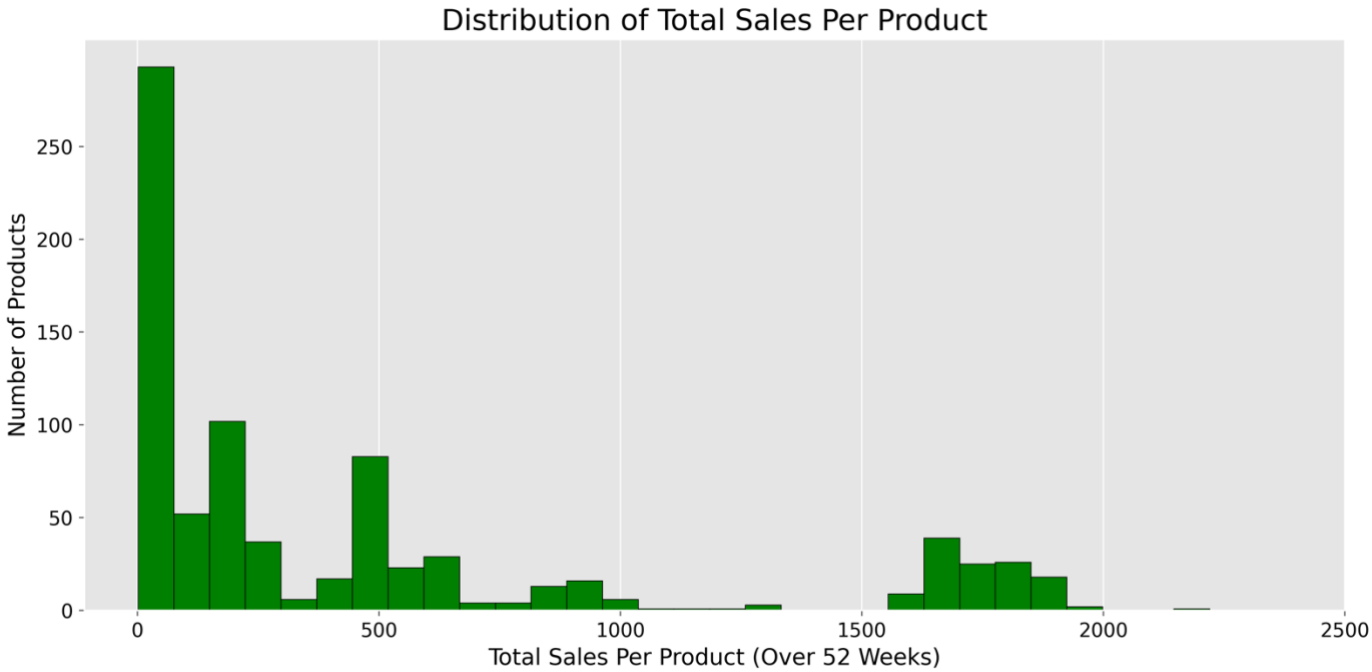
Appendix 2: Histogram 1

This histogram “Total Product Sales Per Week” shows weekly sales for all products throughout the year(W0 to W51).



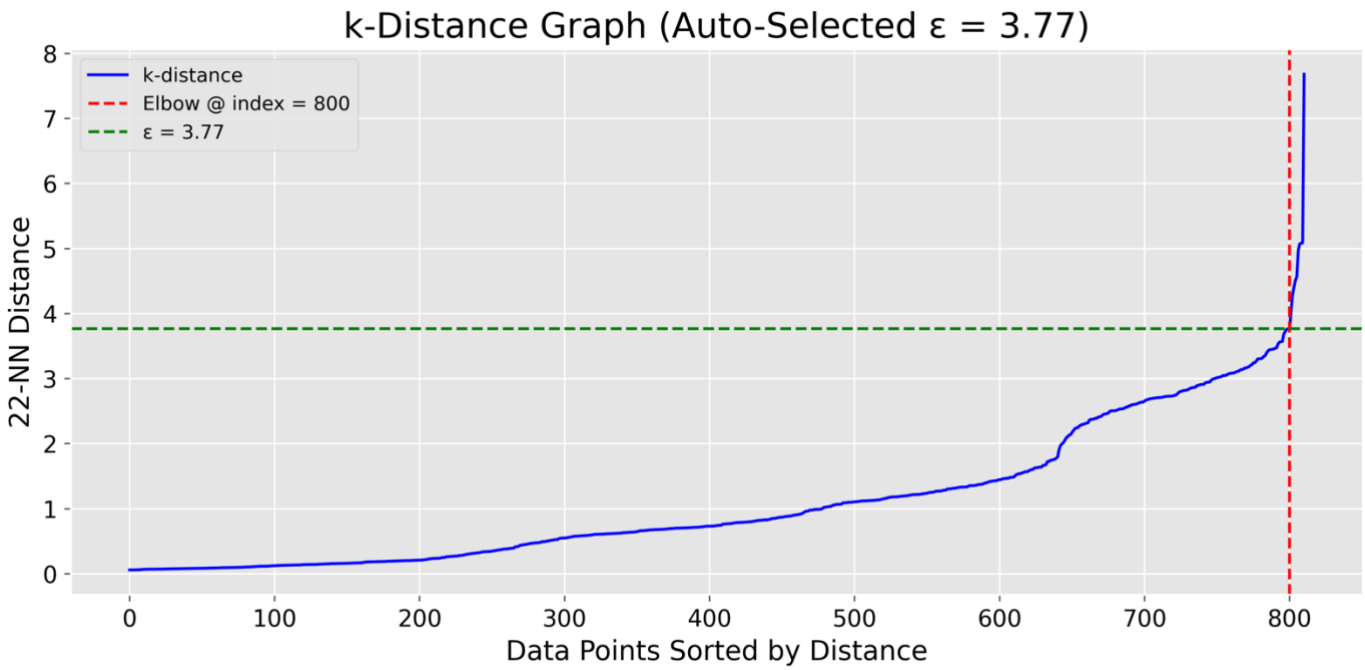
Appendix 3: Histogram 2

This histogram "Distribution of Total Sales Per Product" shows total sales for different products over 52 weeks.



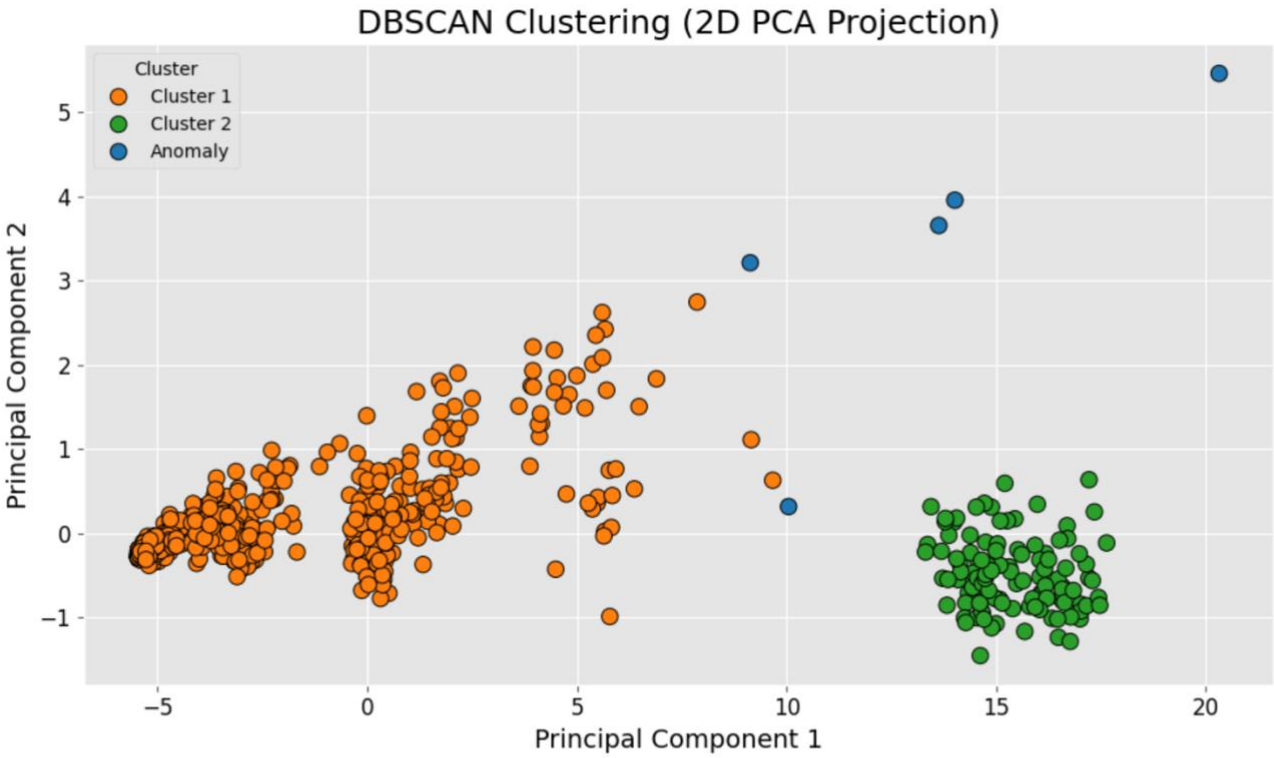
Appendix 4: k-Distance Graph

ε=3.77 automatically selected through k-distance graph.



Appendix 5: DBSCAN Clustering

The clustering illustrates two distinct groups: Cluster 1 and Cluster 2 along with anomalous products exhibiting irregular weekly sales patterns.




```
Cluster Distribution:
Cluster_Name
Anomaly        5
Cluster 1    689
Cluster 2    117
Name: count, dtype: int64

Top 5 Most Anomalous Products:
Product_Code       W0       W1       W2       W3       W4
        P208 2.250000 1.916667 2.750000 2.230769 2.230769
        P262 1.833333 2.333333 2.500000 1.461538 2.615385
        P409 3.250000 3.750000 2.916667 3.000000 2.384615
        P557 2.083333 1.666667 2.666667 2.307692 2.230769
        P615 1.166667 1.083333 1.333333 1.230769 1.153846
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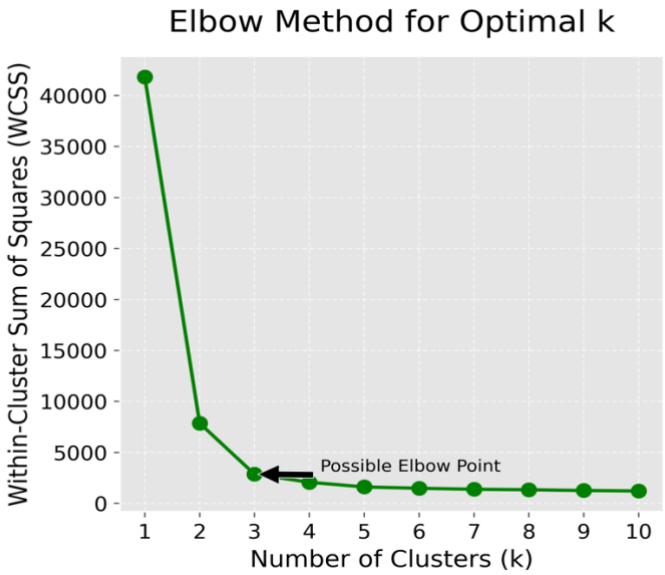
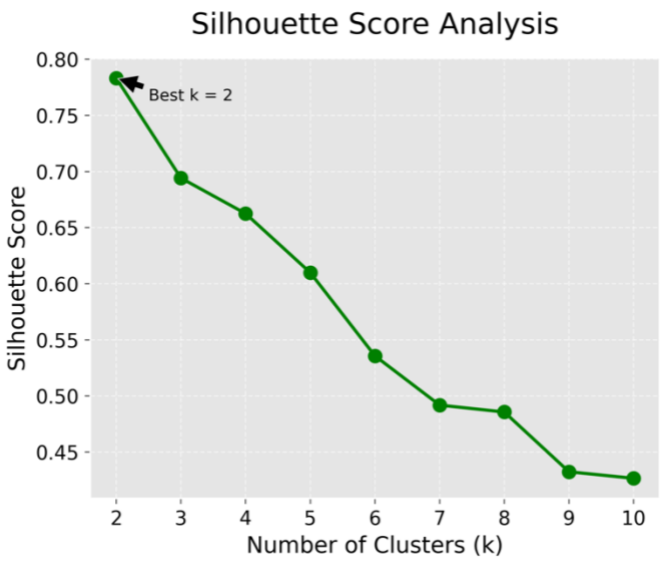
Appendix 6: Elbow Method

The Elbow Method suggested an optimal cluster number of k=3.



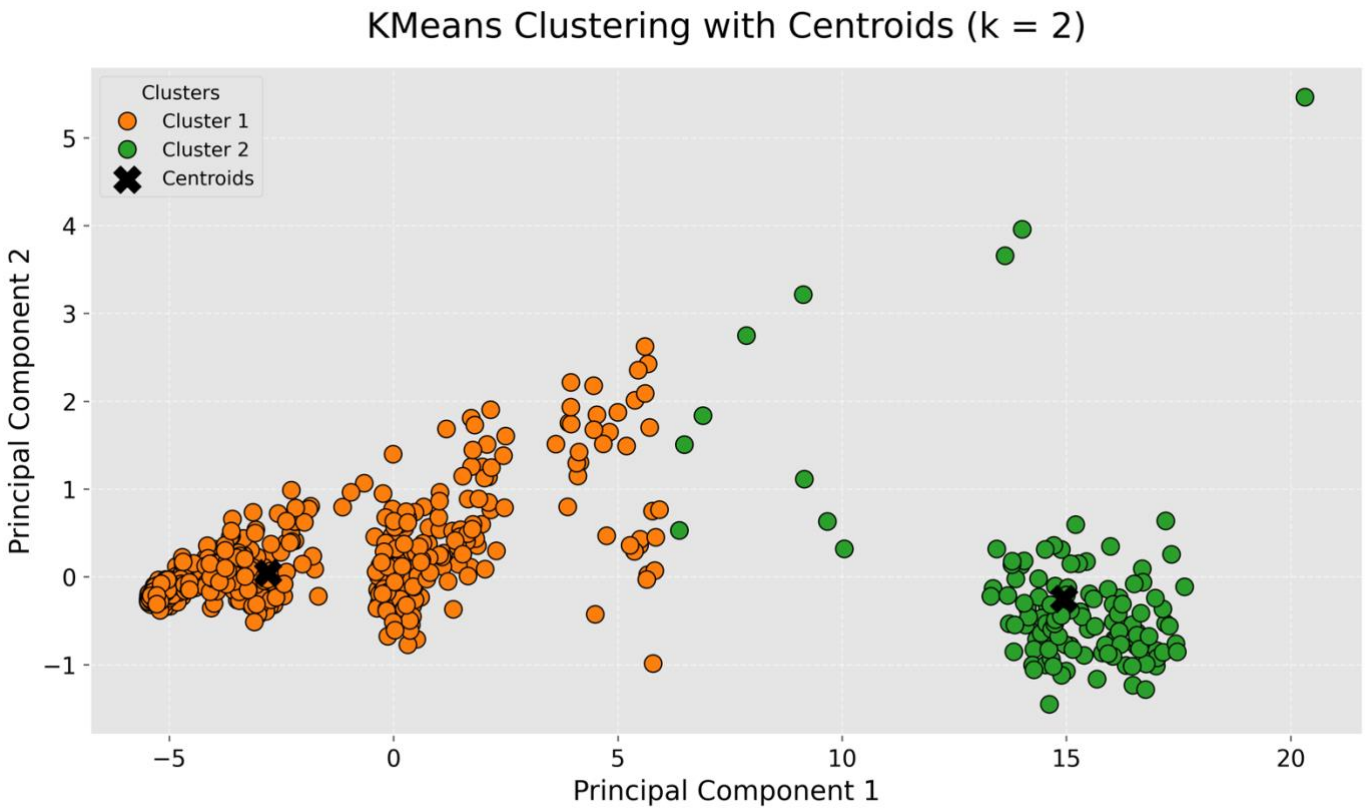
Appendix 7: Silhouette Score

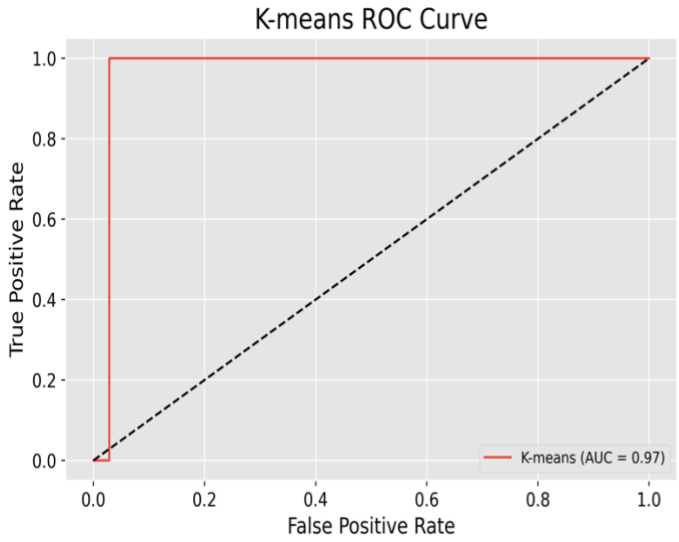
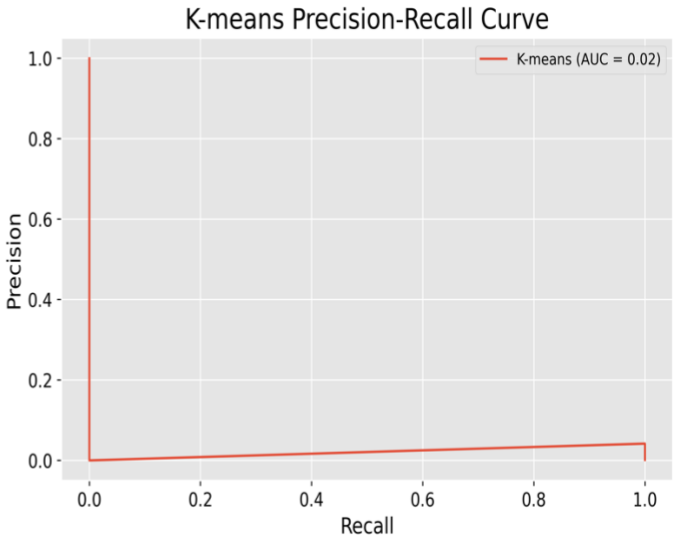
The silhouette score plot peaked at k = 2 with a score of 0.783, indicating that two clusters are optimal for the data.



Appendix 8: KMeans Clustering with Centroids (k = 2)

The clustering plot showed two distinct groups: Cluster 1 and Cluster 2 along with centroids.



Appendix 9: ROC Curve and Precision–Recall Curve

The K-means ROC curve shows strong classification performance (AUC=0.97), while its precision-recall curve reveals poor anomaly detection capability (AUC=0.02), demonstrating the algorithm's high false-positive rate despite good overall accuracy.